



Talgil API behavior guidelines.

Let us start from the description of the Development process.

Before you start the process, it would be nice if you can tell us more about your project and your goals by the connector creation.

Beside that we need to know the following:

1. Which exactly data do you need to get from our system?
2. How do you plan to use the data?
3. How often do you need to update it and make requests?
4. What is the controller ID which you have access to if you have one?
5. Do you own controllers which you plan to work with in the future?

When we get this required information, we will be able to prepare for you our API Development Kit which will include the following:

- Our latest API documentation.
- Access to our Development Server via API and Talgil applications.
- The selected controller Simulator with the configuration of the selected accessible controller.
- Our support during the Development process.

When you have our API Development Kit you will be able to start the Development process on your side.

If you want, we can schedule a meeting where we can help you to find the best endpoint to get the data in the most effective way and will explain to you when it is the best possible time to send this request to get the required data. It can be done by meeting or via Email.

When you will finish the Development process on your side, we will ask you to do the “48-hours load test” to see if everything is working fine and all requirements from our side are fulfilled. To pass it we will ask you to do the following:

- Please run your API connector and leave it running for at least 48 hours.
- In case if you plan to request the historical data by connection of the new controller - please do the separate simulation of the historical request.



The main point of the test is to see if you're holding on to our behavior guidelines and limitations which you will find later in this document and your requests are not dangerous to our Main server and will not damage it in one or other way.

When the test will be passed you will get access to our Production server and real controllers. There you should run only the approved from us requests. In case if you will need to check something or try another request or its parameters you still have access to our Development server where you can do it without any limits.

In case you will want to add new request to the real controllers you will need to pass the test again for new requests and in case if on your side it will be done not in the same job - you will have to test all your request jobs together before it will be approved to be used in the Production server.

If you have a single job which unites all requests it should be tested after each change you made.

If you are not the controller owner and even if you are working on the controller owner behalf you will need to provide filled and signed from the controller owner API Access Permission document.

Now let's talk about features and limitations of our REST API.

We have basically two ways of work with our API:

1. **Realtime.** To get live data directly from our controllers. This way is good when you want to get sensors data, water meter stand, valves status etc... into your external system or your own dashboard. In this case you will get data that actual at the moment of your request.
2. **Database.** To get data from our server database. It is more suitable for analytics and graph building. In this case you will work only with the saved data with some delay. Here you can define the time range of requested data and sampling rate like hourly, daily, monthly.

All database requests containing timestamps "from" and "until" in the request parameters.

In addition, we have several requirements about your requests:

1. **ALL requests should be sent in HTTPS**

The HTTP requests will be blocked. Please be aware of it.

2. **To implement delays after each of your API requests according to the type of the request.** It applies also for requests inside of one burst including the last one in the burst. Please keep in mind that we are working with irrigation and it is no point to bombard our server with several requests in the second. We are living in the other world and here we are counting time in minutes and seconds but not in milliseconds as you probably used to.

Here is the full table of required delays according to its type:

Request type	Minimum interval (s)
Live POST requests (except for modification)	1 second
Live POST modification requests (the old one for the single parameter)	2 seconds
Live POST batch modification requests (ModifyFields/ModifyMany)	15 seconds
Live GET requests	1 second
DB GET log requests (except programs and events) (/log?)	15 seconds
DB GET event log requests (/log?)	15 seconds
DB GET program log requests (/log?)	15 seconds
DB GET water consumption requests (/wc/)	60 seconds
DB GET fert consumption requests (/fc/)	60 seconds

3. **Historical request of the data should be a separate process and has nothing to do with your regular database request.**

When you are connecting the new controller and want to get the data for the long previous period it should be a separate process which should be tested and called separately from your daily activity.



4. **Maximum range for the database requests according to the “rate” parameter.** Unfortunately, we have to limit the accessible time range for the Database requests according to the “rate” parameter. The limit will be set as following:
Hourly = 7 days.
Daily = 31 days.
Weekly = 184 days.
Monthly = 366 days.
Yearly = 731 days.
None = 92 days.
5. **Agree frequency of your requests especially if you will work with Realtime.** Usually, we recommend to request information in Realtime mode not often then every 10-15-20 minutes. If you want to do it more often, please contact us first.
6. **In case you need only specific parameters from different containers by the Real Time requests – please consider to use filters.** More information you will find in the API documentation in **Chapter 8**. It will help to keep the traffic low and to get only data which you really need.
7. **Do not request unnecessary information.** Especially if you work with the Database. We update information in the Database every hour from the minute 19 till the minute 30.
8. **Do not double your requests.** You do not need to send your requests two or more times. The server got your request so give it a bit time to respond.

In case you will find something strange or you need something specific for your needs – please do not hesitate to contact our engineers under our [general mail](#) or our API specialist directly:

Kosta Kiselov

Application engineer

kosta@talgil.com

+972-54-6610189

(WhatsApp on the same number)



Thank you for your attention and we hope for a successful collaboration.
Have a nice day.
Your Talgil service team.